

# Principales matchers de Jasmine

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## 1. not

Invert the matcher following this `expect`

### Example

```
expect(thing).not.toEqual(realThing);
```

## 2. toBeCloseTo(expected, precision)

`expect` the actual value to be within a specified precision of the expected value.

### Parameters:

| Name                   | Type   | Attributes | Default | Description                            |
|------------------------|--------|------------|---------|----------------------------------------|
| <code>expected</code>  | Object |            |         | The expected value to compare against. |
| <code>precision</code> | Number | <optional> | 2       | The number of decimal points to check. |

### Example

```
expect(number).toBeCloseTo(42.213, 3); // Comprueba si number  
redondeado a 3 decimales coincide con 42.213
```

### 3. toBeDefined()

**expect** the actual value to be defined. (Not `undefined`)

#### Example

```
expect(result).toBeDefined();
```

### 4. toBeFalse()

**expect** the actual value to be `false`.

#### Example

```
expect(result).toBeFalse();
```

### 5. toBeTrue()

**expect** the actual value to be `true`.

#### Example

```
expect(result).toBeTrue();
```

### 6. toBeGreaterThan(expected)

**expect** the actual value to be greater than the expected value.

#### Parameters:

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|          |        |                               |
|----------|--------|-------------------------------|
| expected | Number | The value to compare against. |
|----------|--------|-------------------------------|

### Example

```
expect(result).toBeGreaterThan(3);
```

## 7. toBeGreaterThanOrEqual(expected)

`expect` the actual value to be greater than or equal to the expected value.

### Parameters:

| Name     | Type   | Description                            |
|----------|--------|----------------------------------------|
| expected | Number | The expected value to compare against. |

### Example

```
expect(result).toBeGreaterThanOrEqual(25);
```

## 8. toBeInstanceOf(expected)

`expect` the actual to be an instance of the expected class

### Parameters:

| Name     | Type   | Description                                    |
|----------|--------|------------------------------------------------|
| expected | Object | The class or constructor function to check for |

### Example

```
expect('foo').toBeInstanceOf(String);
```

```
expect(3).toBeInstanceOf(Number);
```

```
expect(new Error()).toBeInstanceOf(Error);  
expect(bonoloto()).toBeInstanceOf(Array);
```

## 9. toBeLessThan(expected)

`expect` the actual value to be less than the expected value.

### Parameters:

| Name     | Type   | Description                            |
|----------|--------|----------------------------------------|
| expected | Number | The expected value to compare against. |

### Example

```
expect(result).toBeLessThan(0);
```

## 10. toBeLessThanOrEqual(expected)

`expect` the actual value to be less than or equal to the expected value.

### Parameters:

| Name     | Type   | Description                            |
|----------|--------|----------------------------------------|
| expected | Number | The expected value to compare against. |

### Example

```
expect(result).toBeLessThanOrEqual(123);
```

## 11. toBeNaN()

**expect** the actual value to be `NaN` (Not a Number).

### Example

```
expect(thing).toBeNaN();
```

## 12. toBeNull()

**expect** the actual value to be `null`.

### Example

```
expect(result).toBeNull();
```

## 13. toBeUndefined()

**expect** the actual value to be `undefined`.

### Example

```
expect(result).toBeUndefined();
```

## 14. toContain(expected)

**expect** the actual value to contain a specific value.

### Parameters:

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|          |        |                        |
|----------|--------|------------------------|
| expected | Object | The value to look for. |
|----------|--------|------------------------|

### Example

```
expect(array).toContain(anElement);
expect(string).toContain(substring);
```

## 15. toEqual(expected)

**expect** the actual value to be equal to the expected, using deep equality comparison.

### Parameters:

| Name     | Type   | Description    |
|----------|--------|----------------|
| expected | Object | Expected value |

### Example

```
expect(bigObject).toEqual({ "foo": ['bar', 'baz'] });
```

## 16. toHaveSize(expected)

**expect** the actual size to be equal to the expected, using array-like length or object keys size.

### Parameters:

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

| expected | Object | Expected size |
|----------|--------|---------------|
|----------|--------|---------------|

### Example

```
array = [1,2];

expect(array).toHaveLength(2);
```

## 17. toThrowError(message)

**expect** a function to **throw an Error**.

### Ejemplo

Pondremos como ejemplo una clase Jarra que tiene un setter llamado cantidad el cual lanza un error cuando le pasamos un valor negativo:

```
class Jarra {
  constructor(capacidad, cantidad) {
    this._capacidad = capacidad;
    this._cantidad = cantidad;
  }

  set cantidad(c) {
    if (c < 0) throw new Error('La cantidad debe ser un número positivo');
    this._cantidad = Math.min(c, this._capacidad);
  }
}
```

Desde Jasmine podemos testear que efectivamente se lance el error con un código similar al siguiente:

```
expect(function() { jarra1.cantidad = -5
}).toThrowError('La cantidad debe ser un número positivo');
```